

STATISTICAL CONSULTING IN THE UNIVERSITY: A FACULTY MEMBER'S PERSPECTIVE

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ABSTRACT. Statistical consultants share many common roles and problems. This article addresses some of the issues faced by the educational statistician working as a faculty member in a university. The university milieu, with its emphasis on research, teaching, and service presents some unique opportunities as well as problems for the statistical consultant. The roles, functions, and responsibilities of the educational statistician are emphasized.

A number of articles and several books have been written about statistical or research consulting. In an early bibliography on statistical consulting, Woodward and Schucany (1977) listed 37 references that dealt with different aspects of consulting and with the teaching of consulting. By 1982 the list had almost doubled as Grimes (1982) included 71 references in her updated bibliography. A look through the bibliographies indicates that the literature can be catalogued in a number of ways; articles deal with the teaching and preparation of consultants for jobs in different areas, the role and responsibilities of the statistician, the clientele served, personal characteristics of a good consultant, and organization or administrative issues. A review of the literature also clearly reveals the importance of the consulting role for the statistician regardless of the particular area of expertise. That is, statistical consulting occurs in widely divergent settings.

Although consultants all face common roles and problems, this article addresses some of the issues faced by the educational statistician working as a faculty member in a university. The modifier “educational” indicates that the discussion is centered around those working in the behavioral sciences or Schools of Education as opposed to those working in mathematics/statistics departments, biostatisticians, and so on. Some people use the term “applied statistician” but “educational” is preferred here because it indicates the area of application that is of most interest to this article.

In discussing the functions or objectives of a university, research, teaching, and service are three areas that are almost always mentioned. Similarly, these are also key elements in trying to define faculty workload, criteria for promotion and tenure, and salary increases. How these three areas are defined

and stressed at the educational statistician's university will have a direct bearing on the role that consulting will play in a faculty member's life.

By its very nature, the person working as an educational statistician has a consulting role to play in the university. This role is an integral part of research, teaching, and service and should be recognized and rewarded accordingly.

Role and Function of the Statistical Consultant

As a discipline, educational statistics has a consulting role built into it. It is hard to imagine educational statisticians working in a university where they do not spend a good part of their time consulting with colleagues, students, staff, or outside organizations. In describing the statistician as a jack-of-all-trades and master of one, Barnett (1976) said,

Thus the function of statistics is to provide insight, through (broadly speaking) the analysis of data, on a wide-ranging set of problems of highly associated origin. It is a practical discipline which must be viewed as having a service function for almost any subject area we think of. (p. 263)

As statisticians working in education it is practically impossible to avoid the applied part of the job, and anyone who tries should seriously consider other employment. It is the challenge of problem solving over a wide range of problems in the practice of statistics in education that makes the job so rewarding. Working in a university gives us the opportunity to be masters of our statistical trade, but also jack of many others. As Healy (1973) so aptly put it, "... the statistician is the last of the generalists, the last man for whom knowledge of a little about many things rather than a lot about one thing only is a possible way of life" (p. 72). As we will see later, the person who chooses this "way of life" may also face some additional problems in a university setting.

Several authors have specifically addressed the definition of the statistical consultant as well as the roles and responsibilities that go with the name. I believe they are all consistent with the function of statistics mentioned by Barnett (1976), but they stress different aspects. Joiner (1982) defines statistical consulting in a very broad sense to mean "one or more statisticians working together with one or more persons who need statistical assistance to help solve some problem of interest" (p. 147). He then argues that the type of help given varies over a wide continuum for different types of problems and different clientele, and this determines the quality of the consulting, not whether or not a person is consulting.

Hunter (1981) also characterizes statisticians as working on problems dealing with the collection and analysis of data and that they assume the role of a helper, colleague, or leader, depending on the interaction between the client and the statistician. When both the client and the statistician are actively

involved in the research project, then the statistician is functioning as a colleague. Some prefer to call this a collaborator function. If the client is active and the statistician is passive, then the statistician is a helper and may serve only as a technician. That is, the statistician who may ask a few questions, suggest an analysis, or arrange to have a few computer runs done is not deeply involved. When the client is passive and the statistician is active, then the leader role is operating. Here, the statistician takes the responsibility for all the analysis because the client generally asks, "I have these data, what do they mean?"

In discussing statistical consulting in industry, Marquardt (1979, 1981) argues strongly that total involvement in the consulting situation is the responsibility of the statistician. He lists the following 10 key elements in total involvement (1981, p. 217):

1. Meet clients regularly on their home ground.
2. Learn client's subject and lingo.
3. Understand client's view of the problem.
4. Devise a statistically sound plan of action.
5. Obtain agreement of all parties on plan of action.
6. Monitor progress and expedite if necessary.
7. Analyze the data.
8. Document work and report to management.
9. Follow up until recommendations are implemented.
10. Identify next series of problems to be solved.

With total involvement, the statistician will play many roles, including "problem identifier, idea generator, model builder, collaborator, expeditor, peace-maker, author, politician, and above all, results getter" (1979, p. 103).

Another interesting perspective to the role of and responsibilities of the consultant is given by Feinstein (1970). His comments are addressed to biostatisticians and he talks about the statistician as the "researcher's doctor" and uses the clinician-patient relationship as an analogy to the statistician-consultee relationship. Along with the authority of being a consultant, one must also accept the responsibility. There are plenty of opportunities for "statistical malpractice," and statisticians must become involved in the consulting situation to fulfill their responsibilities. The statistician must stop "regarding his consultative work as an opportunity to disseminate statistical faith among the heathen" (p. 912).

We may not consider ourselves "doctors" in educational research, but some of the people we work with act as if we were. Just as with doctors, some consultees pay us "gullible homage" and treat our pronouncements as "unquestioned omniscience." Of course, on the other extreme, some treat us as witch-doctors whose incantations have no proven value, but tradition dictates that we are to be consulted anyway. To the extent that both the statistician and

the client are actively and equally involved in the situation, neither extreme should occur.

My own view is that, as practicing statisticians in education, we must be involved in consulting and that this involves much more than dispensing statistical formulae. We may do this occasionally, but we should try to minimize our role as passive “helpers” in Hunter’s (1981) terminology. In education, we must be willing to be involved in the whole research process from problem identification and definition to instrument development, research design, statistical analysis, and, finally, the interpretation and writing.

Clientele

Although total involvement in the consulting situation is certainly appealing, it does not apply to all of the situations the educational statistician faces on a day-to-day basis. As we “consult” with other faculty members, staff, and students we run into many less-than-ideal situations. Joiner (1982) argues that while many of these encounters may be bad consulting they still involve the statistician as a consultant. Hyams (1969, 1971) and Sprent (1970) give lists of stereotypical consultees that reflect the wide range of people and situations that may be encountered. A few of these are worth mentioning; they stress some of the problems involved in working in a university.

The “probablist or least significant difference expert” is looking for p -values. They may be led by misguided editors or astrologers who are looking for stars to attach to their results, that is, *, **, and *** for $p < .05$, .01, and .001, respectively. For Salsburg (1973), this involves a “ritualistic invocation of the editorial gods” (p. 153). I am always concerned when consultees smile broadly when the computer yields “ $p < .049$,” but frown deeply when it gives “ $p < .053$.” They may not remember from their introductory statistics class what a Type I error is, but they certainly know that $\alpha = .05$ means “significant.” These consultees often view the role of the statistician as a priest who sprinkles holy water to sanctify their results. As teachers of statistics, we have failed these people when our instruction has stressed only the “significance function” of statistics. Service courses should give more than cursory attention to the total role that statistics plays in data analysis. Cookbook approaches to data manipulation are not what researchers need.

The “numbers collector” comes to you with a mountain of data that is full of jewels or is a gold mine of information. Your job is analogous to the prospector. All you have to do is find the mother-lode and you will be rich. Gehan (1980) warns the consultant to avoid such people because the data are usually collected without regard to hypotheses or general plans. The consultant usually spends a lot of time mining useless information. If you can get the data set on the computer you may have fun playing with it, but don’t expect it to lead to serious scientific research. As Yates (1966) points out, there is a

big difference between data processing and data analysis. Easy access to computer packages is no substitute for rational thought.

The “sporadic leech” is a third type mentioned by Hyams (1971). They are usually a casual acquaintance, or people who got your name from Professor X. They stop you in the hall or call just before lunch to ask you what can be done with their data. These people almost never acknowledge that they needed your help, and they do not really want to share their ideas with you or meet with you to discuss them in any detail. Of course, if the referees question the analysis you are the first to be called, but if not you will never know when the study was published. Kimball (1957) argues that this type of consulting by remote control often leads to errors of the third kind, that is, giving the right answer to the wrong question. Deming (1965) also warns about the dangers of giving informal advice. He says that “a statistician who tries to be a good fellow and give advice under adverse circumstances is in practice and has a client, whether he intended it so or not; and he will later on find himself accountable for the advice” (pp. 1984–1985). Let us hope that when we make errors of the third kind in these consulting situations our names are never mentioned in a footnote. It is certainly embarrassing to find our colleagues using an article as an example of bad research when we are listed in a footnote as the statistical consultant.

Closely related to the sporadic leech is the “5-minuter.” This is usually a person you do not know who was referred to you by one of your colleagues. This type wants to come by immediately or tomorrow at the latest because he or she only needs 5 minutes of your time. Gehan (1980) again cautions the consultant to be wary because the investigator has usually spent a lot of time trying to solve the problem but cannot. Gehan usually finds that the person does not understand the problem and expects you to figure it out and provide solutions, or that there are basic problems or biases in the data that the investigator does not know how to handle. This is a negative example of the “leader” role mentioned by Hunter (1981)—it is your responsibility to bring order out of chaos without the investigator’s help. In any case, you are given little credit for anything you do because “it only took 5 minutes of your time.”

The “amateur statistician” is usually very proficient in making everything fit into one of the models or techniques he or she knows. However, occasionally this person cannot seem to make it fit or has run across a new technique in a statistical package. In these cases, your technical assistance is requested. Without knowing what the person is doing or why, we may be able to answer particular questions, but it is clear that the statistician is a “helper” and the client is the activist who is running the show. Depending on the statistical skill of the client, this may involve “good” consulting but does not always.

The last stereotype mentioned by Hyams (1971) is the “long distance runner.” This person is usually the director of a large grant and wants to hire you

to work for the project. As director, the person has little time to work with you and you are left with the rest of the staff trying to find something—anything—to say. Hyams says these types are running as fast as they can because right behind them is a troupe of horrible demons—their mistakes—waiting to pull them down. Because you are a paid member of the staff, you are expected to provide draft copies of any results found. These are then incorporated into the papers and reports written by the director. No matter what your involvement in the final draft, your name is not mentioned; it is common practice for the staff to remain anonymous.

These stereotypes are oversimplified and do not always lead to bad consulting. However, such situations are ripe with opportunities for statistical malpractices. As professional educational statisticians, we have certain responsibilities that cannot be discharged lightly. Recently the American Statistical Association (ASA) published an interim code of conduct (Ireland et al., 1981) for statisticians. The purpose of these general guidelines is “to provide ethical standards for the professional conduct of its members and to protect individuals and organizations from the misuse of statistics and statistical data” (p. 11). To adhere to these standards is difficult if not impossible when you have only minimal contact with a research project.

At the core of the university is the production and dissemination of knowledge, that is, research and teaching. Since the consulting role can incorporate both of these in giving service to others, the educational statistician should allocate the time necessary to do effective consulting. In the next section, we will see that this is not always the case. But first, there is a special type of client that the educational statistician meets in the university—the student doing a thesis or dissertation.

When the student is your own advisee, some form of total involvement is usually built in. It is difficult, but not impossible, for advisors to avoid their responsibility of working with the student from problem formulation to final revisions. When the students are someone else’s advisees, the whole gamut of involvement is possible. If you teach the service courses in introductory research methods or statistics, then you generally become the reference person when the students in those courses begin their research projects. Many of these students will ask you to be a member of their committee; others will try to use you as a free consulting service.

It is easy in the university to be overwhelmed with dissertation committees. Yet I have not found any administrator willing to set guidelines for what is reasonable. In talking with a number of professors in different areas I have found those who argue that four or five committees are the maximum number that a professor can effectively serve on. Others have agreed that this would be nice, but because of “supply and demand,” they are serving on 20, or in some cases, even as many as 50 committees. It should be obvious that anyone

serving on 15–20 committees with anything approaching total involvement is spending a lot of time each week and most of the evenings toward the end of the semester reading dissertations. It is hard to imagine what the person on 50 committees is doing.

Although you do not have to be an expert in each student's area, some general understanding of the field is important. Mead (1976) points out that a "detailed specialist understanding is of virtually no benefit to the statistician since he is not going to be allowed to use it" (p. 217). After all, when was the last time you heard of a faculty member being promoted or given a large salary increase for serving on a large number of dissertation committees? This point is often not recognized by students.

Mead (1976) suggests that you develop a series of questions to try to precisely define the problem and the experimental units. Lurie (1958, p. 57) suggested earlier that any researcher has the responsibility to answer the following three questions:

1. With respect to the experiment you are performing, just what are your ideas?
2. With respect to the scientific area to which these ideas refer, just what are they about?
3. How sure do you want to be of the correctness of these ideas?

These questions reflect his point of view that the scientist "performs experiments and/or makes observations to obtain data relating to *an idea he has* about the organization of *that portion of the world he is interested in*, so that he can decide *whether his idea was correct or not*" (pp. 57–58).

If the student can intelligently answer the first two questions, then I usually find it personally rewarding to help him answer the third. I believe that because the PhD is supposed to be a research degree, the educational statistician on these committees has the opportunity and, I would argue, the responsibility to exert considerable educational influence on the student concerning the whole research process. For future researchers in education, the dissertation process can serve as a model for what research involves. This influence is especially needed when major advisors abdicate their responsibilities and try to make the methods and results chapters of the dissertation the statistician's responsibility. Students are usually willing to follow their advisor's lead and wait for you to tell them what to do. Given the press of time and other commitments, this is often the easiest thing to do.

Of course, the acquisition of scientific literacy by the student is not the sole responsibility of the educational statistician. By the time students reach the dissertation stage, they should have some basic insights about the structure, collection, inspection, and analysis of data in their field. In some departments this will be true, whereas for others it is not. Some students approach the

dissertation stage with little or no idea of what constitutes research in their field. They will be stumped by Lurie's first questions. Although we can inveigh against this, we will not eliminate it; providing individual tutorials through the dissertation is time consuming and requires a certain missionary zeal that most of us do not have. For those doctoral students who will never do any research as part of their professional responsibilities and whose only "research" will be the dissertation, minimal help may be all that is warranted.

However, regardless of whether we are working with students or faculty members, I think we must take seriously Feinstein's (1970) charge that "A good statistical consultant should not allow clinicians to escape the scientific defects of their own intellectual inertia" (p. 905). As professors we do not necessarily do students or other researchers a favor by providing them with simple answers to complex problems. Feinstein drives the point home when he says, "A clinician who has not yet developed the intellectual 'muscles' that will enable him to 'walk' scientifically will never develop this skill if he constantly avoids the necessary effort by being pushed in a statistical 'wheelchair'" (p. 905).

The Role of Consulting in Teaching, Research, and Service

In identifying the roles, functions, and clientele of the consultant, it is clear that teaching, research, and service are involved. However, consulting has different roles and presents different problems for each of these in the university milieu.

Teaching

It may not be easy to evaluate but at least it is easy to document the teaching load when it is defined in terms of courses taught. However, the teaching aspects of your time spent with colleagues and students in formal and informal consulting sessions is much harder to document. This may explain, in part, why many faculty members have chosen a "closed-door" policy. Yet, Mead (1976) argues that making these contacts, which improve the understanding of statistics in the university, should be an overriding principle by which we operate.

Consulting certainly gives teachers a chance to practice what they preach. The consultant is "likely to find himself continually being forced by his customers up against the edge of his competence" (Healy, 1973, p. 74). The teacher/consultant who does not keep pace with new developments in the statistical field is soon left by the wayside. Two advantages of a university appointment should be the time and the close proximity of statistical colleagues to interact with and learn from. In addition, doctoral students should provide a stimulus to explore new areas.

Consulting can also provide the teacher with an unending supply of good and bad examples, as well as data sets, for use in class. It is almost impossible to teach students the when and why of different techniques with the simplified examples given in most textbooks. In discussing Sprent's paper, Finney (1970) says he is "... a little scared to think that we may find ourselves in the position of teaching *applied statistics* as a formal study with little real relation to the *application of statistics*" (p. 148).

A good consultant and a good teacher also share many of the same personality characteristics and skills. Barnett (1976) stresses the consultant's role of being both a translator and a communicator, and says that communication involves mutual interest, contact, and understanding. Although consulting and teaching are complementary, that does not mean they are always recognized as being equally valuable. Some administrators may view consulting within the university as something entirely separate from teaching. In fact, consulting may be expected in addition to all your other duties. Low, Snee, and Nordheim (1982) argue that classroom efforts suffer when consulting is an addition to the normal load. They point out that the granting of release time is one quantification of the importance that administrators place on consulting. Release time also supports an open door policy of meeting with faculty members and students.

Hanumara et al. (1982) argue that when consulting is not rewarded with tenure, rank, and salary, it is the statisticians' responsibility to enlighten the administration about the worth of consulting to the general mission of a university. At the faculty member's level, they stress the importance of building a dossier on consulting work. This should include both formal and informal consulting. Records should be kept in such a way that they document your contribution to the university's mission.

Research

Research is the cornerstone of the university, and although statistical consulting is certainly research oriented, it may not produce the right kind of research.

Marquardt (1979) stresses that statistical consultants are never working on their own problems but always on other people's problems. Total involvement in solving these problems involves collaborative efforts and generally results in joint authorship. ASA's Committee on Training of Statisticians for Industry says that "... this type of joint publication is indicative of an effective statistician and should be rewarded accordingly" (Ireland et al., 1981, p. 72). In commenting on the report, Box (1980) notes that junior faculty may be judged by those who have "no understanding, or sympathy for, applications" and that those publications "with any taint of application will be down-weighted, if not

entirely discounted” (p. 76). Although Box was talking about statistics departments, educational statisticians may face the same problems.

Low et al. (1982) point out the importance of understanding departmental and university promotion criteria. They suggest that if pure statistical research is necessary, then you wait until after tenure to engage in applied research. However, if applied research is not valued for tenure, then it will not be valued for salary increases or further promotion. I believe the best advice is that you have to be true to our own ideals. If you find the challenge of working with different people to solve diverse problems in several fields intellectually exciting and personally rewarding, then this is what you should do. If you relish the idea of being a generalist who knows about many things, then collaborative research is for you. It must be stressed, however, that collaborative research means the persons must do their homework in the fields they are working in. For this person to try and conform to the “traditional” research mold is not very productive.

Consulting may also provide the educational statistician with methodological problems that have no standard answers. These unsolved problems can provide the same kind of challenge and excitement for some people as content problems do for others. Methodological problems are more central to statistics as a discipline and generally fit into the “traditional” mode that the authors mentioned above are talking about.

Box (1976) stresses the importance of a closed feedback loop between theory and practice. When there is an open cycle that is stuck in either the practice mode or the theory mode, then stagnation can occur. He describes the maladies that may result as either “cookbookery” or “mathemistry.” The symptoms of cookbookery are “. . . a tendency to force all problems into the molds of one or two routine techniques, insufficient thought being given to the real objectives of the investigation or to the relevance of the assumptions implied by the imposed methods” (p. 797). On the other hand, “. . . mathemistry is characterized by development of theory for theory’s sake which, since it seldom touches down with practice, has a tendency to redefine the problem rather than solve it” (p. 797).

Educational statisticians who only practice cookbookery in their consulting roles are probably not doing scholarly research. However, the person involved in interdisciplinary collaborative research with several colleagues is just as much a scholar as someone who practices mathemistry. Cox (1959) once defined research as a war on ignorance, and I am arguing that the war can be fought on different fronts.

Given a commitment to quality research, it is hard to say no to other faculty members or students needing research help without some feelings of guilt. However, quantity of output should not be substituted for quality. When it becomes obvious that existing commitments cannot be maintained without

doing substandard noninvolved work, then the educational statistician must say no.

In a research-oriented university, you should give your highest priorities to genuinely collaborative projects that result in joint publications; second should come advisory work in which you are consulted about specific design and analysis problems; and third should be situations in which a technician role of doing routine statistical tasks is involved. In selecting colleagues to work with, Daniel (1969) suggests that you look for people who are responsive, instructable, willing, and able. They should also have clearly formulated problems that have some importance to their field.

Service

Service often plays the role of an unwanted stepchild in the functions of the university, at least in the minds of many faculty members and promotion and tenure committees. It is not as clearly defined and not as highly valued as teaching and research.

Minton and Freund (1977) define service as “including consultative assistance in statistics to students and faculty in support of thesis and other research” (p. 114). They argue that the quality of research is enhanced throughout the university when consultative assistance is available as a service to other units. If you agree with Minton and Freund, then educational statisticians in the university can provide considerable service to the school. This type of service goes beyond the collaborative research defined above and includes the more routine day-to-day interactions with faculty and students about their research problems. In these cases, service overlaps with the informal teaching aspects of consulting addressed earlier.

In many universities, however, “service” does not include service within the university, but instead means unpaid external “professional community/public service.” For instance, the University of Pittsburgh excludes from this type of service all instructional activities and all activities for which you are paid directly by an outside agency. At the University of Pittsburgh, the types of activities called service by Minton and Freund (1977) fall under the rubric of “Academic Support Activities,” which includes “providing services that directly assist the academic functions of the institution.”

The opportunity to provide free methodological assistance to educators outside the university certainly exists for educational statisticians. This assistance could even be managed by the department or school the faculty member is in. However, I do not think you will see this type of service accounting for large blocks of time until it is valued on a par with research and teaching. Given the current demands on most faculty members’ time, they will not, as a group, seek opportunities outside the university to work for free. For an excellent discussion of the issues surrounding public service in higher edu-

cation, see the monograph by Crosson (1983). She clearly delineates many of the functions and priorities associated with public service in colleges and universities.

The model employed by some corporations might be tried by the university. That is, corporate executives are sometimes granted paid leave to work on public service projects. The university might grant sabbaticals based on public service as well as those based on research.

Summary

I have argued that educational statisticians working in a university department should find consulting an integral part of their jobs. Given the very nature of educational statistics, consulting with others is at the core of the field and also gives life to the faculty member's teaching and research. That is, consulting is not something one does in addition to teaching, research, and service. Rather, consulting is part of and complementary to each of these.

However, stressing the consulting role is not without its difficulties. A number of problems have been identified that are associated with the university milieu. These are not new problems or unique to any one school or department. One advantage of reading the extensive literature that has developed around statistical consulting is the reassurance that these are common problems faced by statisticians working in many different fields.

The educational statistician who develops the reputation of being a good consultant is quickly overloaded with faculty, students, and outside clients needing help. However, the time demands for reading and studying, teaching, and research must also be met. In this area, people are left to their own needs and values to set priorities. There are no commonly accepted workload documents that tell faculty members how their time should be allotted. If the person is continually overcommitted, then a frank discussion with the department chairperson or dean about the school's priorities and work expectations is needed. Although many educational statisticians will not want to develop a formal code of conduct as Deming (1965, 1966, 1972) has done, some clear thought must be given to how much help and what type of responsibilities they are willing to assume.

In conclusion, I find statistical consulting an exciting and rewarding part of my job. Although I have experienced many of the frustrations and problems mentioned above, I would not want to reduce or give up this consulting role.

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